

Exp No.7: Analog Beamforming Analysis Using MATLAB

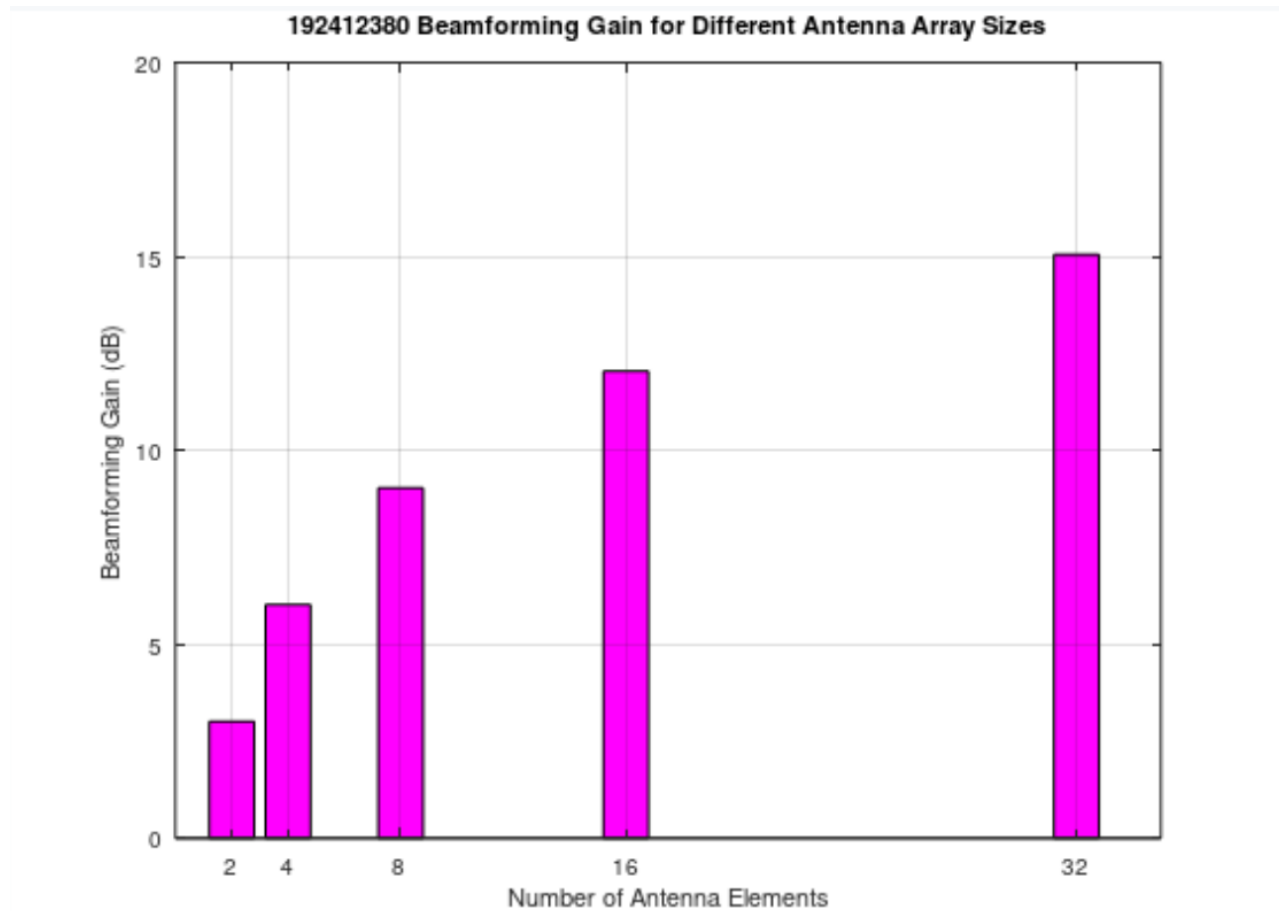
AIM: To analyze the performance of analog beamforming by studying beamforming gain, beam width, and main lobe direction using MATLAB.

Objective 1: To analyze the beamforming gain achieved by varying the number of antenna elements in an analog beamforming system using MATLAB.

Program :

```
1 |clc;
2 clear;
3 close all;
4
5 % Number of Antenna Elements
6 antennas = [2 4 8 16 32];
7
8 % Beamforming Gain (dB)
9 gain = 10 * log10(antennas);
10
11 figure;
12
13 % Bar Graph (Magenta Color)
14 bar(antennas, gain, 'm');
15
16 grid on;
17
18 xlabel('Number of Antenna Elements');
19 ylabel('Beamforming Gain (dB)');
20 title('192412380 Beamforming Gain for Different Antenna Array Sizes');
21
22 drawnow;
23 pause(5);
```

OUTPUT :



Objective 2: To compare the beam width of an antenna array for different beamforming configurations using MATLAB.

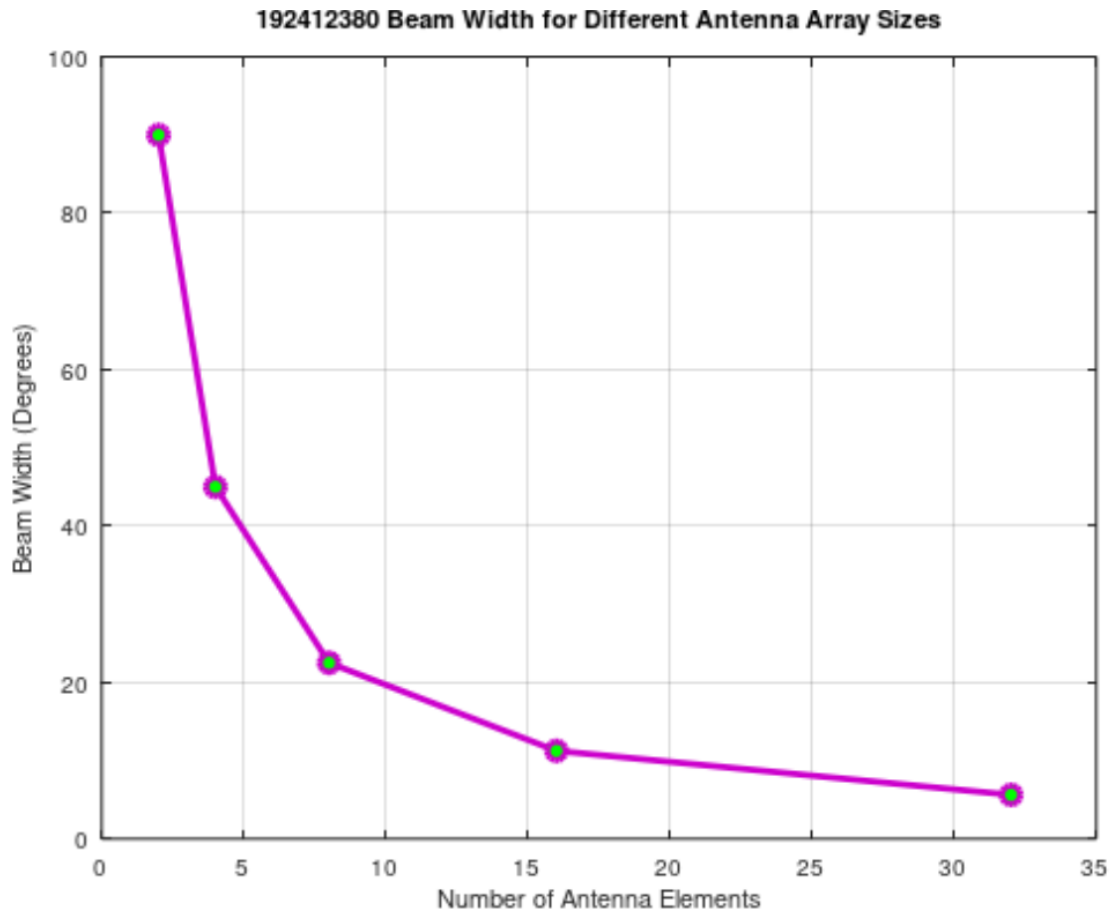
Program

```

1  clc;
2  clear;
3  close all;
4
5  % Number of Antenna Elements
6  antennas = [2 4 8 16 32];
7
8  % Beam Width (Degrees)
9  beamwidth = [90 45 22.5 11.25 5.625];
10
11 figure;
12
13 % Plot with Different Colors
14 plot(antennas, beamwidth, ...
15      '-o', ...
16      'Color', [0.8 0 0.8], ...      % Purple line
17      'MarkerFaceColor', [0 1 0], ... % Green markers
18      'LineWidth', 2);
19
20 grid on;
21
22 xlabel('Number of Antenna Elements');
23 ylabel('Beam Width (Degrees)');
24 title('192412380 Beam Width for Different Antenna Array Sizes');
25
26 drawnow;
27 pause(5);

```

OUTPUT :

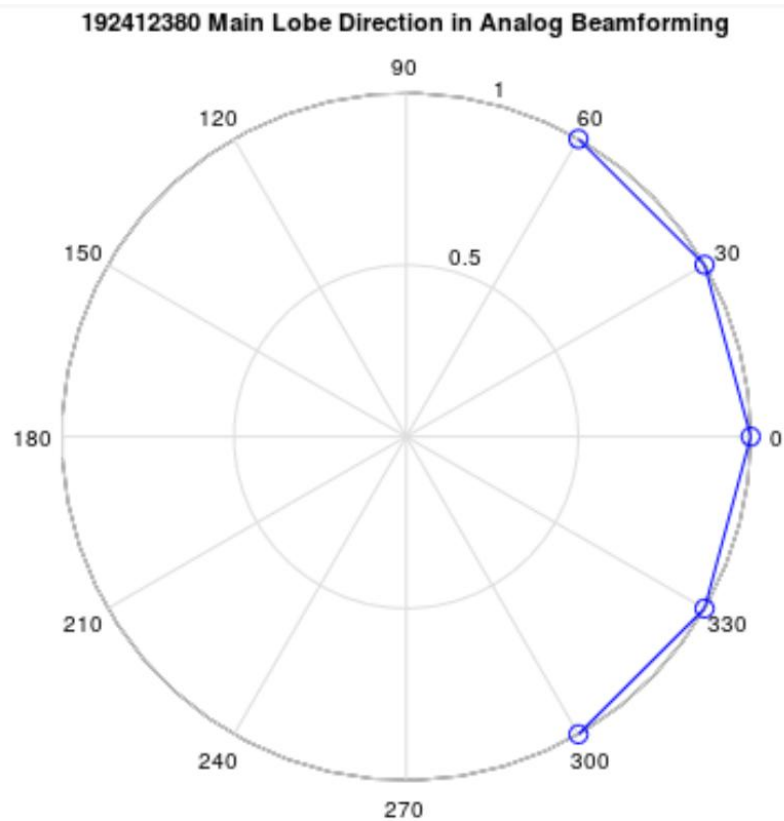


Objective 3: To analyze the main lobe direction by steering the antenna array beam to different angles using MATLAB.

program

```
1 clc;
2 clear;
3 close all;
4
5 theta = deg2rad([-60 -30 0 30 60]);
6 r = [1 1 1 1 1];
7
8 figure;
9 polar(theta, r, 'bo-');
10 title('192412380 Main Lobe Direction in Analog Beamforming');
```

OUTPUT :



Result:

The MATLAB analysis successfully demonstrated the characteristics of analog beamforming. The results showed that increasing the number of antenna elements improves beamforming gain and reduces beam width, while phase adjustment enables the main lobe to be steered toward different directions.

Practical Questions

A wireless communication system employs **analog beamforming** to improve signal strength and coverage by directing the transmitted beam toward a desired user location using antenna arrays. By adjusting the phase of antenna elements, the beam can be steered, and key performance parameters such as beamforming gain, beam width, and main lobe direction can be analyzed. Using MATLAB, simulate an analog beamforming system and analyze its performance.

1. To analyze the beamforming gain achieved in an analog beamforming system by varying the number of antenna elements and observe its effect on signal strength.
2. To compare the beam width of an antenna array for different antenna configurations and study how beam width changes with increasing antenna elements.
3. To analyze the main lobe direction of the antenna array by steering the beam toward different angles using phase adjustments in an analog beamforming system.